

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Second Semester of M. Tech. Examination (Mechanical, CAD/CAM)

May 2018

ME755/ME707.01 Computer Aided Manufacturing

Date: 08.05.2018, Tuesday

Time: 01:30 p.m. to 04:30 p.m.

Maximum Marks: 70

Instructions:

1. Section I and II must be attempted in separate answer sheets.
2. Make suitable assumptions and draw neat figures wherever required.
3. Use of scientific calculator is allowed.

SECTION- I

Q - 1 Answer the question below (Each carry 1 mark). [05]

- (i). In finish machining of an island on a casting with CNC milling machine, an end mill with 10 mm diameter is employed. The corner points of the island are represented by (0, 0), (0, 30), (50, 30) and (50, 0). By applying cutter radius right compensation, the trajectory of the cutter will be

- (a) (-5, 0), (-5, 35), (55, 35), (55,-5), (-5,-5)
- (b) (0,-5), (55,-5), (55, 35), (-5, 35), (-5,-5)
- (c) (5, 5), (5, 25), (45, 25), (45, 5), (5, 5)
- (d) (5, 5), (45, 5), (45, 25), (5, 25), (5, 5)

- (ii). In an NC machining operation, the tool has to be moved from point (5,4) to point (7,2) along a circular path with center at (5,2). The correct G and M code for this motion is

- (a) N010 G02 X7.0 Y2.0 I0.0 J-2.0
- (b) N010 G03 X7.0 Y2.0 I5.0 J2.0
- (c) N010 G01 X7.0 Y2.0 I5.0 J2.0
- (d) N010 G00 X7.0 Y2.0 I5.0 J2.0

- (iii). The tool of an NC machine has to move along a circular arc from (5, 5) to (10, 10) while performing an operation. The center of the arc is at (10, 5). Which one of the following NC tool path commands performs the above mentioned operation?

- (a) N010 G02 X5 Y5 R5
- (b) N010 G03 X5 Y5 R5
- (c) N010 G01 X10 Y10 R5
- (d) N010 G02 X10 Y10 R5

- (iv). M30 stands for:

- (a) End of program
- (b) End of block
- (c) End of tape and tape rewind
- (d) Coolant on/ off

- (v). In a CNC program block, N002 G02 G91 X40 Z40..., G02 AND G91 refer to

- (a) Circular interpolation in counterclockwise direction and incremental dimension
- (b) Circular interpolation in counterclockwise direction and absolute dimension
- (c) Circular interpolation in clockwise direction and incremental dimension
- (d) Circular interpolation in clockwise direction and absolute dimension

Q - 2(a) Describe with sketch the working and construction of recirculating ball screw used in CNC machine tools stating advantages. What is preloading? [04]

Q - 2(b) Answer any three. [12]

- (i) Explain advance features of CNC programming.

- (ii) What is the importance of using subroutines in a part program? Explain with an example.
- (iii) How are the axes of CNC machine designated? Explain with sketch stating illustration of vertical milling machine.
- (iv) Differentiate between open loop and closed loop control system.
- (v) Explain Geometrical statements of APT stating examples.

Q – 3 Write a manual part program for part as shown in **figure 1** (All dimensions are in mm) without using cutter radius compensation. The thickness of part is 10 mm. Assume 10 mm cutter diameter.

[14]

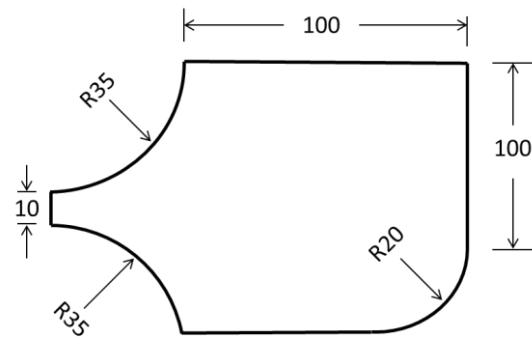


Figure 1

OR

Q – 3 Write a manual part program for part shown in **figure 2** (All dimensions are in mm). Assume only finished cut is required.

[14]

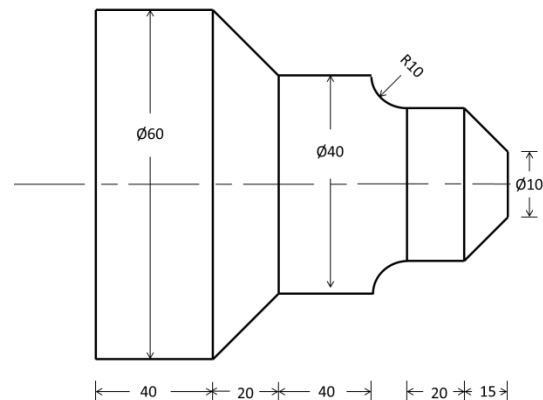


Figure 2

SECTION- II

- Q - 4 (a)** What do you mean by computer aided inspection? Explain the steps in vision processing system. **[04]**
- (b)** Discuss the significance of rapid prototyping in current manufacturing scenario. **[03]**

- Q - 5 (a)** Explain CODE system used for part classification and coding of group technology. [06]
(b) Describe types of GT machine cell in brief. [04]
(c) Explain process of fused deposition modeling technique of RP. [04]

OR

- Q - 5 (a)** Define flexibility in context of FMS. Explain various types of flexibility. [06]
(b) State types of FMS layout and explain open field type layout with suitable diagram. [04]
(c) Explain major elements of coordinate measuring machines. [04]

- Q - 6 (a)** Apply the Rank Order Clustering (ROC) technique to the part-machine incidence matrix in the following table to identify logical part families and machine groups. [08]

Machines	Parts						
	1	2	3	4	5	6	7
1	1	1				1	1
2			1	1	1		
3	1					1	1
4						1	1
5			1	1			

- (b)** What is CIM wheel? Draw and explain CIM wheel. [06]

OR

- Q - 6 (a)** Describe computer integrated manufacturing (CIM) in detail. Illustrate the different elements of CIM. [06]
(b) Explain the aspects that one should consider in implementing CIM. [04]
(c) Discuss major components of flexible manufacturing system. [04]